

QUANG-HUY NGUYEN

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RESEARCH INTERESTS

Computational Biology: My work spans multi-omics studies integrating genomics, transcriptomics, metabolomics, and proteomics. This includes bulk, single-cell, and spatially resolved analyses, pathway analysis, variant calling (both germline and somatic), genome-wide association studies, and polygenic risk scores.

Computational Healthcare Informatics: I develop and apply data-driven methods for learning from heterogeneous clinical and biological data to support decision-making in medicine and public health.

I also address core AI challenges raised by these problems, including multi-modal modeling, agentic reasoning, uncertainty quantification, federated and collaborative learning, etc.

EDUCATION

Auburn University

Ph.D. in Computer Science & Software Engr - cGPA: 3.73/4.0

Advisor: Professor Wei-Shinn Ku.

Auburn, AL

08/2023 – Present

- Developed *FedeKD*, a federated knowledge distillation framework for robust knowledge transfer under heterogeneous settings. Achieved a significant reduction in negative transfer while improving predictive performance across 6 vision and medical imaging datasets.
- Developed *scBatchProx*, a federated learning-inspired framework for post-hoc single-cell batch integration via latent refinement, enabling continual and cumulative updates under dataset growth. Achieved Macro-F1 up to 0.76 with robustness to extreme batch imbalance.
- Developed *FedWQ-CP*, a federated conformal prediction framework for uncertainty quantification under joint data and model heterogeneity. Achieved robust client-level and global coverage across 7 vision and medical imaging datasets.

Dai Nam University

Bachelor of Pharmacy - cGPA: 8.09/10.00

Hanoi, Vietnam

10/2015 – 08/2020

PUBLICATIONS

†co-first author, *co-corresponding author

Preprints & Manuscripts Under Review

5. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *scBatchProx: Federated-Inspired Refinement for Stable Cell-Type Discriminability under Heterogeneous Batch Compositions*. **arXiv**, 2602.00423. ([Under review](#)). 2026
4. **Quang-Huy Nguyen**, Zeru Zhang, Duc-Hau Le*, Jake Y. Chen*, Wei-Shinn Ku*, Hao Chen*, Zongliang Yue*. *PAG-Agent: a biologist-oriented research assistant for context-aware pathway-level analysis and interpretation*. **bioRxiv**, 2026.06.02.729674. ([Under review](#)). 2026
3. Hoang Le†, Van-Minh Nguyen†, Duc-Hai Tran, Van-Anh Tran, Van-Nui Nguyen, **Quang-Huy Nguyen**†*, Duc-Hau Le*. *A Semi-Supervised Deep Learning Framework for Multi-Omics Cancer Subtype Classification and Biomarker Discovery*. **bioRxiv**, 2022.01.13.476268. ([Under review](#)). 2026
2. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *Conformalized Neural Networks for Federated Uncertainty Quantification under Dual Heterogeneity*. **arXiv**, 2602.23296. ([Under review](#)). 2026
1. **Quang-Huy Nguyen**, Jiaqi Wang*, Wei-Shinn Ku*. *FedeKD: Energy-Based Gating for Robust Federated Knowledge Distillation under Heterogeneous Settings*. **arXiv**, 2605.05553. ([Under review](#)). 2026

Peer-reviewed publications

10. Trang T. H. Tran, ..., **Quang-Huy Nguyen**, ..., Quan Nguyen*, Van H. Vu*, Nam S. Vo*. *VN1K is a pangenome-informed multi-omics and phenomics resource for the Vietnamese population*. **Nature Communications**, 17:6679. 2026
9. Ching-Lan Chang†, Michael A. Mosh†, **Quang-Huy Nguyen**†, ..., David Hess*, Edwin C. Oh*. *Wastewater intelligence predicts the emergence of clinically-relevant and drug-resistant *Candidozyma auris* at healthcare facilities*. **Nature Communications**, 17:5415. 2026
8. **Quang-Huy Nguyen**, Ha Nguyen, Edwin C. Oh, Tin Nguyen. *Current approaches and outstanding challenges of functional annotation of metabolites: a comprehensive review*. **Briefings in Bioinformatics**, 25(6):bbae498. 2024

7. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *DrGA: cancer driver gene analysis in a simpler manner*. **BMC Bioinformatics**, 23:86. 2022
6. **Quang-Huy Nguyen**, Tin Nguyen, Duc-Hau Le. *Identification and validation of a novel three hub long noncoding RNAs with m6A modification signature in low-grade gliomas*. **Frontiers in Molecular Biosciences**, 9:801931. 2022
5. **Quang-Huy Nguyen**, Duc-Hau Le. *oCEM: Automatic detection and analysis of overlapping co-expressed gene modules*. **BMC Genomics**, 23:39. 2022
4. Thi Hai Yen Nguyen†, Tin Nguyen, **Quang-Huy Nguyen†***, Duc-Hau Le*. *Re-identification of patient subgroups in Uveal Melanoma*. **Frontiers in Oncology**, 11:731548. 2021
3. **Quang-Huy Nguyen**, Duc-Hau Le. *Similarity calculation, enrichment analysis, and ontology visualization of biomedical ontologies using UFO*. **Current Protocols in Bioinformatics**, 1:e115. 2021
2. **Quang-Huy Nguyen**, Duc-Hau Le. *Improving existing analysis pipeline to identify and analyze cancer driver genes using multi-omics data*. **Scientific Reports**, 10:20521. 2020
1. **Quang-Huy Nguyen**, Hung Nguyen, Tin Nguyen, Duc-Hau Le. *Multi-omics analysis detects novel prognostic subgroups of breast cancer*. **Frontiers in Genetics**, 11:574661. 2020

PEER-REVIEWED CONFERENCES

1. Hung Nguyen, Bang Tran, Duc Tran, **Quang-Huy Nguyen**, Duc-Hau Le, Tin Nguyen. *Disease subtyping using community detection from consensus networks*. **2020 12th International Conference on Knowledge and Systems Engineering (KSE)**, Can Tho, Vietnam, IEEE, 2020; 318–323. Oral Presentation.

RESEARCH EXPERIENCE

Department of Computer Science, HUST

Visiting Graduate Student

Hanoi, Vietnam

12/2024 – 01/2025

Advisor: Associate Professor Duc-Hau Le.

- Developed *PAG-Agent*, a web-based, LLM-driven platform for context-aware pathway analysis of bulk and single-cell RNA-seq data, unifying differential expression analysis, pathway enrichment, meta-analysis, literature-assisted interpretation, and hypothesis generation. Designed a novel context-aware pathway prioritization framework that incorporates experimental conditions and research objectives to improve the identification of biologically relevant pathways while reducing generic enrichment signals, a common limitation of conventional ORA-based methods.
- Developed *DeepSSC*, a semi-supervised multi-omics classifier for cancer subtype prediction. Achieved 96.0% accuracy (99.2% AUC) across 3 TCGA cohorts, with up to 19% improvement over graph-based baselines; mentored undergraduate students on model development.

Department of Computer Science and Software Engineering, Auburn University

Graduate Research Assistant

Auburn, AL

08/2023 – 08/2024

Advisor: Research Assistant Professor Phi Hung Bya and Associate Professor Tin Nguyen.

- Conducted a literature review on functional analysis methodologies in LC-MS-based targeted and untargeted metabolomics. Dockerized 13 surveyed software tools as services, simplifying installation and execution to enhance accessibility for readers with limited IT expertise.
- Conducted a literature review on large language models and their applications in biomedical domain.
- Led the *Candidozyma auris* project; performed BLAST-based homology mapping and KEGG/GO enrichment analyses to characterize pathways involved in stress adaptation and antifungal resistance; contributed to manuscript editing and proofreading.
- Led the type 2 diabetes (T2D) polygenic score project; developed a genotype-to-pathway analysis pipeline that derives gene-level reference-deviation signals directly from VCF genotypes without matched transcriptomic data, identified T2D-associated pathways, and demonstrated strong predictive performance of pathway-specific polygenic risk scores; co-drafted the manuscript.

School of Computer Science and Engineering, Thuyloi University

Visiting Scientist

Hanoi, Vietnam

10/2021 – 07/2023

Advisor: Associate Professor Duc-Hau Le.

- Developed *DrGA*, an R toolkit for automated analysis of cancer driver genes, unifying gene set enrichment, gene-clinical association, module-level analysis, and patient stratification workflows.

School of Computer Science and Engineering, Thuyloi University

Hanoi, Vietnam

Visiting Student

12/2017 – 07/2019

Advisor: Associate Professor Duc-Hau Le.

- Read and replicated research papers, designed experiments and conducted algorithm on high-dimensional omics data in R.

INDUSTRY EXPERIENCE

Center for Biomedical Informatics, Vingroup Big Data Institute

Hanoi, Vietnam

Research Scientist

10/2020 – 09/2021

Managers: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

- Developed *oCEM*, an R package for detecting overlapping co-expressed gene modules using decomposition methods (ICA/IPCA), with a permutation-based approach for selecting the optimal number of components.
- Designed and implemented a multi-omics analysis pipeline in R that detected novel prognostic subgroups of breast cancer, achieving a Cox log-rank p-value of 0.002 compared to the PAM50 classification scheme's p-value of 0.598. Led manuscript preparation and writing.
- Co-mentored an undergraduate student in clustering uveal melanoma patients in R.
- Contributed to the design and planning of the VN1K variant detection pipeline for large-scale genomic analysis of the Vietnamese population. Contributed to manuscript proofreading.

Research Intern

03/2020 – 10/2020

Internship mentor: Associate Professor Duc-Hau Le.

- Developed and implemented a pipeline in R to systematically identify and analyze cancer driver genes using multi-omics data, ensuring a consistent approach to uncover key genetic factors contributing to cancer progression.

RESEARCH PROJECTS AND COMPETITIONS

nano-scFMs (🔗 Repo)

Auburn, AL

Independent Open-Source Project

06/2026 – Present

- Built a collection of lightweight, faithful PyTorch reimplementations of single-cell foundation models to improve readability, reproducibility, and accessibility for AI biology research.
- Reimplemented model architectures, preprocessing pipelines, tokenization workflows, pretrained checkpoint loading, and embedding extraction for single-cell *.h5ad* data.

Cell Maps for AI (CM4AI) CodeFest, UAB

Birmingham, AL

NIH Bridge2AI Consortium

08/2025

- Collaborated in a team of 5 to create separate embeddings of yeast fitness data, proteomic mass spectrometry, and cellular imaging, then fuse them into a co-embedding to construct a with-yeast multiscale subcellular hierarchy that uncovered hidden biological processes in triple-negative breast cancer.
- Implemented the analysis pipeline based on the agreed workflow in Python.

The 1000 Vietnamese Genomes Project (VN1K)

Hanoi, Vietnam

Vingroup Joint Stock Company

2019 – 2024

PI: Associate Professor Duc-Hau Le and Dr. Nam Sy Vo.

GenomeAsia100K

Hanoi, Vietnam

NTU and others

2017 – 2022

PI: Dr. Stephan Schuster (Singapore side) and Associate Professor Duc-Hau Le (Vietnam side).

HONORS AND AWARDS

Next Gen Leadership Award

Auburn, AL

Issued by AGBT Precision Health

07/2026

- Selected as one of 10 Next Gen Leaders worldwide from a highly competitive pool of applicants across academia, healthcare, and industry for the 2026 AGBT Precision Health Meeting, with an invitation to present PAG-Agent and scholarship support for conference participation.

Benchmarks Resource Grant Program

Auburn, AL

Kaggle Research Grant Program

07/2026

- Awarded a Benchmarks Resource Grant to develop PAG-Agent's open-source benchmark for evaluating LLMs in context-aware biological pathway prioritization, with expanded computational resources provided through Kaggle.

Gold Reviewer Award

Auburn, AL

Issued by ICML 2026

05/2026

- Recognized as a Gold Reviewer at ICML 2026, ranking in the top 25% of eligible reviewers based on Area Chair ratings.

Google PhD Fellowship in Health Research (University-Level Nomination)

Auburn, AL

Associated with Auburn University

04/2026

- Selected as one of three doctoral students nominated university-wide by AU Graduate School for the 2026 Google PhD Fellowship in Health Research.

Certificate of Merit for Academic Excellence and Class Leadership

Hanoi, Vietnam

Associated with Dai Nam University

08/2020

- Top 3% of the Department's 2015 cohort (nearly 110 students).
- Class president in 5 consecutive years (2015-2020).

27th Mathematics Olympic for Students

Nha Trang, Vietnam

Issued by Vietnam Mathematical Association

04/2019

- Received an honorable mention in Linear Algebra among 640 undergraduate competitors representing 80 institutions nationwide.
- Attended as the first and only representative of my university and became the first Vietnamese non-engineering student in the history of the competition.

TEACHING EXPERIENCE

Department of Computer Science and Software Engineering, Auburn University

Auburn, AL

Graduate Teaching Assistant for COMP-1220 Python & COMP-1230 Matlab

08/2024 – Present

- Teaching Python and Matlab fundamentals to ≈ 60 undergraduates and guiding them through problem-solving exercises in lab sections.

PROFESSIONAL SERVICE

- **Conference Reviewer:** ICML 2026, NeurIPS 2026, ICDM 2026, AAAI 2027.
- **Journal Reviewer:** Frontiers in Genetics (2021), Frontiers in Cell and Developmental Biology (2022), Frontiers in Bioinformatics (2026), Frontiers in Artificial Intelligence (2026).

PRESENTATIONS AND TALKS

1. 2024 MW CTR-IN Annual Meeting at The Westin Las Vegas Hotel & Spa, Las Vegas, NV. *Topic: A Systems-level Approach for Predicting Patient Survival using Multi-omics Data.* 2024

TECHNICAL SKILLS

- **Programming:** Python, R, Matlab, JavaScript
- **Single-Cell & Omics:** Scanpy, AnnData, MuData, Seurat
- **Web Development:** HTML, CSS, RESTful APIs
- **Deep Learning & ML:** PyTorch, TensorFlow 2, scikit-learn
- **Database:** MySQL, SQLite, Oracle, MongoDB
- **Clouds & Tools:** BigQuery, AWS, Git, GitHub, Docker